

MADHAVA KRISHNAN A

Public Policy Researcher | Technology, Climate, Environment and Urban Governance
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PROFILE

Electronics and communication engineer turned public policy researcher, with an MPP from IIT Delhi (May 2026) focused on technology governance, climate and environmental policy, and evidence-based policymaking. Brings a rare combination of engineering foundations and interdisciplinary policy training, with hands-on experience spanning qualitative and quantitative research, GIS-based spatial analysis, PESTLE-based strategic foresight, institutional analysis, and policy brief production. Has worked across digital public infrastructure, groundwater governance, urban climate resilience, platform regulation, cooperative federalism, and international development. Equally comfortable with desk-based analytical work, field-based primary research, stakeholder engagement, and policy communication for technical and non-technical audiences. Available immediately for full-time roles in New Delhi.

EDUCATION

- Master of Public Policy (MPP)** May 2026
Indian Institute of Technology Delhi (IIT Delhi), School of Public Policy
Thesis: GIS-based Spatial Multi-Criteria Analysis (SMCA) for Nature-Based Solutions prioritisation in urban flood management, Chennai. Built a multi-criteria indicator framework integrating spatial, hydrological, and socio-economic data layers to produce evidence-based infrastructure planning recommendations.
- Bachelor of Engineering, Electronics and Communication Engineering** 2016 – 2020
Kumaraguru College of Technology (KCT), Coimbatore | Anna University
Coursework and projects in signal processing, embedded systems, IoT systems design, and data communication networks.
- Master of Arts, Urban Development** 2021 – 2024
Indira Gandhi National Open University (IGNOU)

PROFESSIONAL EXPERIENCE

- GIS Research Consultant (Independent)** Nov 2025 – Feb 2026
Article 14, Earth Journalism Network-funded investigation
- Provided GIS analysis for an investigative piece by Makepeace Sitlhou and Greeshma Kuthar on forest cover change in Manipur, published in Article 14 and credited as "With GIS inputs from Madhava Krishnan."
 - Conducted satellite imagery analysis and spatial overlay of poppy cultivation extent with historical forest cover data to assess temporal patterns of deforestation and land use change across the region.
- Policy Research Intern** May – Jul 2025
WELL Labs (Water, Environment, Land and Livelihoods Lab), Bengaluru
- Conducted in-depth policy analysis on groundwater governance, mapping inter-scheme coordination across three centrally sponsored schemes (Jal Jeevan Mission, AMRUT, PMKSY) and identifying convergence opportunities and implementation bottlenecks.
 - Produced policy briefs, issue notes, and strategy documents synthesising primary field research with government data and academic literature, for internal stakeholders and government partners.
 - Designed institutional frameworks with indicator-based monitoring and evaluation pathways for integrating para-hydrologist networks into Karnataka's groundwater management system.
 - Conducted 2 field visits to Chintamani district and facilitated 1 focus group discussion with ward residents to gather primary data on last-mile groundwater service delivery.
 - Tracked national policy and regulatory developments across urban and rural water governance programs, synthesising trends into briefings for program leadership.
- R&D Intern** Jan – Jun 2020
Hoffensoft Technologies, Coimbatore
- Designed and built an IoT-based environmental monitoring prototype covering sensors, firmware, and a real-time data analytics dashboard, gaining hands-on experience in technology development and field data pipelines.
 - Conducted basin-level field data collection and coordinated cross-functional teams on prototype deployment and iterative testing.
- Maintenance Engineer Trainee** Jun 2020 – May 2021
Pasteur Institute of India (Government of India Undertaking), Coonoor

- Managed technology systems operations and regulatory compliance in a Government of India R&D establishment, producing compliance reports and operational logs for government audits.

RESEARCH AND SELECTED PROJECTS

Climate, Environment and Urban Resilience

- Urban Resilience through Sponge Parks (Master's Thesis, IIT Delhi): GIS-based SMCA framework using weighted, multi-criteria indicators to prioritise Nature-Based Solutions (NbS) infrastructure sites across Chennai. Integrated spatial, socio-economic, and hydrological data to produce evidence-based planning recommendations for urban flood management.
- Landslide Susceptibility Modelling, Uttarkashi: Multi-layer GIS risk zonation combining topographic, hydrological, and land-use indicators to generate a quantitative susceptibility index for climate-driven geohazard risk.
- Chennai Floods Impact Assessment: Assessed climate-driven flood risk using spatial and socio-economic data, examining vulnerability of urban communities and infrastructure to extreme precipitation events.

Strategic Foresight and Technology Governance

- Food Security Futures 2050: PESTLE-based strategic foresight exercise examining long-term trajectories of climate change, technological disruption, and geopolitical shifts on global food systems. Developed scenario pathways and policy recommendations across a 25-year horizon, covering international governance, trade, and resource security dimensions.
- ONDC Digital Commerce and Platform Governance Study: Mixed-methods primary research on India's Open Network for Digital Commerce (ONDC) using semi-structured interviews with two MSMEs (Aruvi Eco Organics and Bhooshan Enterprises). Assessed the Beckn protocol as a model of Digital Public Infrastructure (DPI), with analysis of platform governance, algorithmic accountability, equitable access, and technology sovereignty implications.
- DESMAP Spatial Equity Framework: GIS-based analysis assessing equity in digital and physical infrastructure allocation using multi-layer spatial indicators for participatory planning.

Governance, Institutions and Social Policy

- NITI Aayog Institutional Reform Strategy: Analysed cooperative federalism structures and inter-governmental coordination mechanisms; produced recommendations for decentralised planning reform benchmarked against international institutional models.
- Delhi Water Crisis Policy Analysis: Mapped the stakeholder ecosystem across public utilities, informal sector actors, and regulatory bodies, assessing governance failures using supply-demand and political economy frameworks.
- CBI Institutional Drift Analysis: Applied Mahoney-Thelen theory of gradual institutional change to examine veto-player dynamics, accountability gaps, and reform pathways in India's premier investigative agency.
- Bonded Labour Recruitment Study: Regulatory gap analysis benchmarking India's legal framework against international standards, with cross-sectoral implications for labour and inclusive development policy.
- Katwaria Sarai Community Governance Study: Multi-method fieldwork using household surveys and stakeholder interviews to produce evidence-based community governance recommendations.

COMPETENCIES AND SKILLS

Research Methods: qualitative methods (focus group discussions, semi-structured interviews, ethnography, document analysis, thematic coding), quantitative methods (multi-criteria analysis, indicator frameworks, SMCA, PESTLE scenario planning), mixed methods design, Python for data analysis

Spatial Data and GIS: ArcGIS and QGIS for spatial analysis and mapping, multi-layer indicator-based frameworks, DEM and hydrological analysis, land-use and socio-economic data integration, risk zonation and infrastructure prioritisation

Strategic Foresight: PESTLE-based scenario development and trend analysis, long-horizon policy forecasting, international benchmarking and peer-country comparison, technology trajectory and ecosystem mapping

Policy Writing and Communication: policy briefs, research reports, issue notes, strategy documents, concept notes, technical memos, and presentations for government, research, and public audiences

Domain Areas: water and environmental governance, climate resilience and urban flooding, digital public infrastructure and platform regulation, technology governance and AI policy, cooperative federalism and institutional reform, labour and social policy, food and resource security, urban development

Tools: Python (data analysis and visualisation), ArcGIS, QGIS, Microsoft Office Suite, data synthesis from multi-source and large datasets

LANGUAGES

English (Fluent, professional working proficiency) | Tamil (Native) | Hindi (Working proficiency)